

SET THEORY

BASIC CONCEPTS

SET: A set is a collection of well defined objects.

ELEMENTS OR MEMBERS: Objects constituting the set are called elements or members of the set.

$x \in A \Rightarrow x$ belongs to A

$x \notin A \Rightarrow x$ does not belong to A .

SET NOTATION

$$N = \{1, 2, 3, \dots\}$$

In set notation, we write this as

$$N = \{x : x \text{ is a natural number}\}$$

We use the following alphabets to represent certain sets.

A = set of vowels = $\{a, e, i, o, u\}$

N = set of natural numbers = $\{1, 2, 3, \dots\}$

W = set of whole numbers = $\{0, 1, 2, 3, \dots\}$

Z = set of integers = $\{0, \pm 1, \pm 2, \dots\}$

$I = Z_+$ = set of positive integers = $\{1, 2, 3, \dots\}$

Q = set of rational numbers

R = set of real numbers

C = set of complex numbers

FINITE AND INFINITE SETS: A set is said to be finite if it contains finite number of distinct elements. Otherwise it is infinite.

Eg.

Finite - set of vowels, set of days of a week

Infinite - set of even integers

NULL SET: A set having no element is called a null set or empty set. It is denoted by $\{\}$ or ϕ .

Eg.

$\{x \in I / x^2 + 1 = 0\}$ is a null set.

UNIVERSAL SET: Universal set is the set of all elements under discussion.

SINGLETON SET: A set having exactly one element is called singleton set

Eg.

$$A = \{x/x-1=0\} = \{1\}$$

CARDINALITY: Let A be a finite set. Then the number of different elements in A is called the cardinality of A and is denoted by $|A|$ or $n(A)$ or $\#A$.

SUBSET: Let A and B be any two sets. We say that A is a subset of B if every element of A is an element of B . We denote it as $A \subseteq B$. Here B is called the superset of A and is denoted by $B \supseteq A$.

Eg.

If $A = \{2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5\}$ then clearly $A \subseteq B$.

$$\text{Also } N \subseteq W \subseteq Z \subseteq R \subseteq C$$

IMPROPER SUBSET: Given a set A , it is obvious that $A \subseteq A$ and $\phi \subseteq A$. Both ϕ and A are called improper subsets of A .

PROPER SUBSET: For two sets A and B , if $B \subseteq A$ and $A \neq B$ then A is called a proper subset of B . It is denoted by $A \subset B$.
i.e. $A \subset B \Leftrightarrow (A \subseteq B \text{ and } A \neq B)$

POWER SET: Given a set A , the set consisting of all the subsets of A is called the power set of A . It is denoted by $P(A)$ or 2^A .

Note that if a finite set A has n elements, then the power set of A namely $P(A)$ has 2^n elements.

Eg.

$$A = \{a, b\}$$

$$P(A) = \{\phi, \{a, b\}, \{a\}, \{b\}\}$$

EQUALITY: Two sets A and B are equal iff every element of A is an element of B and every element of B is an element of A .

of A. i.e. $A \subseteq B$ and $B \subseteq A$.

Eg.

$$A = \{0, 1\} \text{ and } B = \{x/x(x-1) = 0\} \text{ then } A = B.$$

Note:

(1) If $A = \{1, 3, 5\}$ and $B = \{5, 1, 3\}$ then $A = B$.

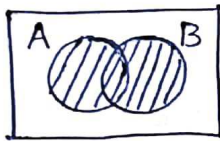
(2) If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 3, 4, 2\}$ then $A = B$.

SOME OPERATIONS ON SETS

* UNION OF SETS

The union of any two sets A and B written as $A \cup B$ is the set of all elements which are in A or B or both.

$$A \cup B = \{x/x \in A \text{ or } x \in B\}$$



Eg.

(1) $A = \{2, 5, 8\}$ $B = \{1, 9, 10\}$

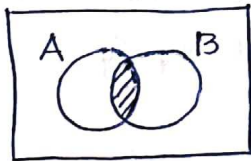
then $A \cup B = \{1, 2, 5, 8, 9, 10\}$

(2) $A \cup \phi = A$ (3) $\mathbb{Z} \cup \mathbb{Q} = \mathbb{Q}$

* INTERSECTION OF SETS

The intersection of any two sets A and B written as $A \cap B$ is the set of all elements which are in both A and B.

$$A \cap B = \{x/x \in A \text{ and } x \in B\}$$



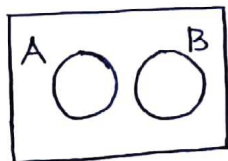
Eg.

$A = \{1, 3, 8\}$ $B = \{4, 5, 8\}$

then $A \cap B = \{8\}$

* DISJOINT SETS

Two sets A and B are said to be disjoint iff $A \cap B = \phi$ i.e. A and B have no elements in common.



Eg.

$A = \{2, 4, 6\}$ $B = \{1, 3, 5\}$

$A \cap B = \phi \Rightarrow A$ and B are disjoint

* DISJOINT COLLECTION

A collection of sets is called a disjoint collection if for every pair of sets in the collection, the 2 sets are disjoint.

The elements of a disjoint collection are said to be mutually disjoint.

Eg.

$$\text{Let } A_1 = \{1, 3, 6\}, A_2 = \{2, 7, 9\}, A_3 = \{4, 5, 8, 10\}$$

$$\text{Note that } A_1 \cap A_2 = \phi, A_2 \cap A_3 = \phi \text{ and } A_1 \cap A_3 = \phi$$

$\therefore A_1, A_2$ and A_3 are mutually disjoint and hence $\{A_1, A_2, A_3\}$ is a disjoint collection.

* COMPLEMENT OF A SET

The complement of a set A denoted by A^c or A' is the set of all elements which are not in A but in the universal set.

$$A^c = \{x \in E / x \notin A\}$$

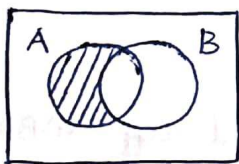
Eg.

$$(1) \text{ Let } E = \{1, 2, 3, 4, 5\} \text{ and } A = \{1, 2, 4\}. \text{ Then } A^c = \{3, 5\}$$

$$(2) \text{ Let } E = \mathbb{R} \text{ Then } \mathbb{Q}^c = \text{set of all irrational numbers.}$$

* DIFFERENCE OF SETS

Let A and B be any two sets. Then the relative complement of B with respect to A denoted by $A - B$ is the set of all elements in A but not in B . It is also denoted by A/B . i.e. $A - B = \{x / x \in A \text{ and } x \notin B\}$



Eg.

$$A = \{2, 4, 6, 8, 10\} \quad B = \{1, 2, 3, \dots, 10\}$$

$$\text{Then } A - B = \{2\} \text{ and } B - A = \{1, 3, 5, 7, 9\}$$

$$\text{Note that } A - B \neq B - A$$

PROBLEMS

1) List all proper subsets of $\{1, 2, 3\}$

Soln.

$$\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$$

2) Enumerate the elements in the following sets

(1) $\{x \in \mathbb{R} / x^2 - 3x + 2 = 0\}$

(2) $\{x \in \mathbb{R} / x^2 + 1 = 0\}$

(3) $\{x \in \mathbb{C} / x^2 + 9 = 0\}$

Soln.

(1) $x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2 \in \mathbb{R}$

$\therefore \{x \in \mathbb{R} / x^2 - 3x + 2 = 0\} = \{1, 2\}$

(2) $x^2 + 1 = 0 \Rightarrow x = \pm i \notin \mathbb{R}$

$\therefore \{x \in \mathbb{R} / x^2 + 1 = 0\} = \{\}$

(3) $x^2 + 9 = 0 \Rightarrow x = \pm 3i \in \mathbb{C}$

$\therefore \{x \in \mathbb{C} / x^2 + 9 = 0\} = \{3i, -3i\}$

3) Write the improper subsets of $\{a, b, c, d, e\}$

Soln.

ϕ and $\{a, b, c, d, e\}$

4) Determine which of the following are equal - $\phi, \{\phi\}, \{0\}$

Soln.

None are equal since

ϕ - set containing no element

$\{\phi\}$ - singleton set containing empty set as an element

$\{0\}$ - singleton set containing zero as an element

5) What is the power set of empty set? What is the power set of $\{\phi\}$?

Soln.

Let $A = \phi$. Then $P(A) = \{\phi\}$

Let $B = \{\phi\}$. Then $P(B) = \{\phi, \{\phi\}\}$

6) What is the cardinality of the set A if the cardinality of the power set $P(A)$ is 256?

Soln.

Cardinality of $P(A) = 2^n = 256$ (given)

$\Rightarrow n = 8$

∴ Cardinality of A is 7.

7) Find the intersection of the following sets

$$\{x/x^2-1=0\} \text{ and } \{x/x^2+2x+1=0\}$$

Soln.

$$\begin{aligned} & \{x/x^2-1=0\} \cap \{x/x^2+2x+1=0\} \\ &= \{1, -1\} \cap \{-1\} \\ &= \{-1\} \end{aligned}$$

8) If $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{1, 2, 3, 6\}$ what are $\bigcup_{i=1}^3 A_i$ and $\bigcap_{i=1}^3 A_i$?

Soln.

$$\bigcup_{i=1}^3 A_i = A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 6\}$$

$$\bigcap_{i=1}^3 A_i = A_1 \cap A_2 \cap A_3 = \{2\}$$

LAWS OF SET THEORY

1) COMMUTATIVE LAWS

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Proof:

$$\text{Let } x \in A \cup B$$

$$\Leftrightarrow x \in A \text{ or } x \in B$$

$$\Leftrightarrow x \in B \text{ or } x \in A$$

$$\Leftrightarrow x \in B \cup A$$

$$\therefore A \cup B = B \cup A$$

2) ASSOCIATIVE LAWS

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

Proof:

$$\text{Let } x \in A \cup (B \cap C)$$

$$\Leftrightarrow x \in A \text{ or } x \in B \cap C$$

$$\Leftrightarrow x \in A \text{ or } (x \in B \text{ or } x \in C)$$

$$\Leftrightarrow (x \in A \text{ or } x \in B) \text{ or } x \in C$$

$$\Leftrightarrow x \in A \cup B \text{ or } x \in C$$

$$\Leftrightarrow x \in (A \cup B) \cup C$$

$$\therefore A \cup (B \cap C) = (A \cup B) \cap C$$

3) DISTRIBUTIVE LAWS

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Proof:

$$\text{Let } x \in A \cap (B \cup C)$$

$$\Leftrightarrow x \in A \text{ and } x \in B \cup C$$

$$\Leftrightarrow x \in A \text{ and } (x \in B \text{ or } x \in C)$$

$$\Leftrightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$$

$$\Leftrightarrow x \in A \cap B \text{ or } x \in A \cap C$$

$$\Leftrightarrow x \in (A \cap B) \cup (A \cap C)$$

$$\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4) DEMORGAN'S LAWS

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Proof:

$$\text{Let } x \in (A \cup B)^c$$

$$\Leftrightarrow x \notin A \cup B$$

$$\Leftrightarrow x \notin A \text{ and } x \notin B$$

$$\Leftrightarrow x \in A^c \text{ and } x \in B^c$$

$$\Leftrightarrow x \in A^c \cap B^c$$

$$\therefore (A \cup B)^c = A^c \cap B^c.$$

5) IDENTITY LAWS

$$A \cup \phi = \phi \cup A = A$$

$$A \cap \xi = \xi \cap A = A$$

6) COMPLEMENT LAWS

$$A \cup A^c = \xi$$

$$A \cap A^c = \phi$$

7) IDEMPOTENT LAWS

$$A \cup A = A$$

$$A \cap A = A$$

8) NULL LAWS

$$A \cup \xi = \xi$$

$$A \cap \phi = \phi$$

9) ABSORPTION LAWS

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

10) INVOLUTION LAW

$$(A^c)^c = A$$

PROBLEMS BASED ON LAWS

- 1) Prove that (i) $A - B = A \cap B^c$
(ii) $A \subseteq B \Leftrightarrow B^c \subseteq A^c$

Solution

$$(i) \text{ Let } x \in A - B$$

$$\Leftrightarrow x \in A \text{ and } x \notin B$$

$$\Leftrightarrow x \in A \text{ and } x \in B^c$$

$$\Leftrightarrow x \in A \cap B^c$$

$$\therefore A - B = A \cap B^c$$

$$(ii) A \subseteq B \Leftrightarrow x \in A \Rightarrow x \in B$$

$$\Leftrightarrow x \notin B \Rightarrow x \notin A$$

$$\Leftrightarrow x \in B^c \Rightarrow x \in A^c$$

$$\Leftrightarrow B^c \subseteq A^c$$

- 2) Let A, B, C be 3 sets and if $A \subseteq B$ and $B \cap C = \phi$, prove that $A \cap C = \phi$

Solution

Assume the contrary i.e. assume $A \cap C \neq \phi$.

⇒ There exists an element x such that

$$x \in A \cap C$$

$$\Rightarrow x \in A \text{ and } x \in C$$

$$\Rightarrow x \in B \text{ and } x \in C \quad (\because A \subseteq B)$$

$$\Rightarrow x \in B \cap C$$

$$\Rightarrow x \in \phi \quad (\text{given } B \cap C = \phi)$$

which is a contradiction since no element can belong to a null set.

∴ Our assumption is wrong.

$$\text{Hence } A \cap C = \phi.$$

3) Prove that $(A-B) \cap (B-A) = \phi$

Solution

$$\text{LHS} = (A-B) \cap (B-A)$$

$$= (A \cap B^c) \cap (B \cap A^c)$$

$$= (A \cap A^c) \cap (B \cap B^c)$$

$$= \phi \cap \phi$$

$$= \phi = \text{RHS}$$

4) Prove that for any 2 sets A and B , $A - (A \cap B) = A - B$.

Solution

$$\text{LHS} = A - (A \cap B)$$

$$= A \cap (A \cap B)^c$$

$$= A \cap (A^c \cup B^c)$$

$$= (A \cap A^c) \cup (A \cap B^c)$$

$$= \phi \cup (A - B)$$

$$= A - B = \text{RHS}$$

5) Prove that $A - (B \cup C) = (A-B) \cap (A-C)$

Solution

$$\text{RHS} = (A-B) \cap (A-C)$$

$$= (A \cap B^c) \cap (A \cap C^c)$$

$$= (A \cap A) \cap (B^c \cap C^c)$$

$$= A \cap (B \cup C)^c$$

$$= A - (B \cup C)$$

$$= \text{LHS}$$

6) Prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$

Solution

$$\begin{aligned}(A \cap B) - (A \cap C) &= (A \cap B) \cap (A \cap C)^c \\&= (A \cap B) \cap (A^c \cup C^c) \\&= (A \cap B \cap A^c) \cup (A \cap B \cap C^c) \\&= (\emptyset \cap B) \cup [A \cap (B \cap C^c)] \\&= \emptyset \cup [A \cap (B - C)] \\&= A \cap (B - C)\end{aligned}$$

7) If A, B and C are three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$ then prove that $B = C$.

Solution

To prove that $B \subseteq C$ and $C \subseteq B$.

Let $x \in B$

$$\Rightarrow x \in A \cup B \quad (\because B \subseteq A \cup B)$$

$$\Rightarrow x \in A \cup C \quad (\because A \cup B = A \cup C)$$

$$x \in B \Rightarrow x \in A \text{ or } x \in C \quad \text{--- ①}$$

If $x \in A$, then $x \in A$ and $x \in B$

$$\Rightarrow x \in A \cap B$$

$$\Rightarrow x \in A \cap C \quad (\because A \cap B = A \cap C)$$

$$x \in A \Rightarrow x \in A \text{ and } x \in C$$

$$\Rightarrow x \in C \quad \text{--- ②}$$

Substitute ② in ①

$$x \in B \Rightarrow x \in C \text{ or } x \in C$$

$$\Rightarrow x \in C$$

$\therefore B \subseteq C$. Similarly we can prove that $C \subseteq B$.

Hence $B = C$.

PARTITION OF A SET

Consider a set of students. This set could be partitioned into subsets where each subset contains students in same row (say).

This should be such that

- (1) The sum of the students in all the subsets should be equal to the total number of students
- (2) Each student should be in only one subset
- (3) Each subset should be nonempty i.e. there should be atleast one student in each row.

Then the set of all these subsets forms a partition of the given set of students.

DEFINITION

Let A be a nonempty set. Then any set of subsets $A_1, A_2, A_3, \dots$ of A will form a partition of A if

- (1) $A_1 \cup A_2 \cup A_3 \cup \dots = A$
- (2) The subsets A_i are all mutually disjoint
i.e. $A_i \cap A_j = \phi$ for $i \neq j$
- (3) Each $A_i \neq \phi$

Eg.

Let $A = \{a, b, c, d\}$ and let $A_1 = \{b\}$, $A_2 = \{a, c\}$ and $A_3 = \{d\}$ be subsets of A . Then $\{A_1, A_2, A_3\}$ is a partition of A since

- (1) $A_1 \cup A_2 \cup A_3 = A$
- (2) $A_1 \cap A_2 = \phi$, $A_1 \cap A_3 = \phi$ and $A_2 \cap A_3 = \phi$
- (3) $A_1 \neq \phi$, $A_2 \neq \phi$ and $A_3 \neq \phi$ i.e. A_1, A_2 and A_3 are nonempty

*Note that there can be more than one partition for a set.

PROBLEMS

- 1) Find all the partitions of
 - (i) $\{a, b, c\}$
 - (ii) $\{1, 2, 3, 4\}$

Solution

(i) ① $\{\{a, b, c\}\}$

② $\{\{a\}, \{b\}, \{c\}\}$

③ $\{\{a, b\}, \{c\}\}$

④ $\{\{a, c\}, \{b\}\}$

⑤ $\{\{b, c\}, \{a\}\}$

(ii) ① $\{\{1, 2, 3, 4\}\}$

② $\{\{1\}, \{2\}, \{3\}, \{4\}\}$

③ $\{\{1, 2\}, \{3, 4\}\}$

④ $\{\{1, 3\}, \{2, 4\}\}$

⑤ $\{\{1, 4\}, \{2, 3\}\}$

⑥ $\{\{1, 2\}, \{3\}, \{4\}\}$

⑦ $\{\{1, 3\}, \{2\}, \{4\}\}$

⑧ $\{\{1, 4\}, \{2\}, \{3\}\}$

⑨ $\{\{2, 3\}, \{1\}, \{4\}\}$

⑩ $\{\{2, 4\}, \{1\}, \{3\}\}$

⑪ $\{\{3, 4\}, \{1\}, \{2\}\}$

⑫ $\{\{1\}, \{2, 3, 4\}\}$

⑬ $\{\{2\}, \{1, 3, 4\}\}$

⑭ $\{\{3\}, \{1, 2, 4\}\}$

⑮ $\{\{4\}, \{1, 2, 3\}\}$

2) Let $S = \{1, 2, 3, \dots, 9\}$. Determine whether or not each of the following is a partition of S .

(i) $\{\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}\}$ - No since union does not contain 7.

(ii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}\}$ - No since intersection of the first and third subsets is not empty

(iii) $\{S\}$ - Yes

(iv) $\{\{1, 4, 5, 6, 7\}, \{7\}, \{2, 3, 8, 9\}\}$ - No since each subset has to be nonempty

(v) $\{\{1, 2, 7\}, \{3, 5\}, \{4, 6, 8, 9\}, \{3, 5\}\}$ - Yes

CARTESIAN PRODUCT

Let A and B be any two sets. Then the cartesian product of A and B denoted by $A \times B$ is the set of all ordered pairs where the first element belongs to A and the second element belongs to B .

i.e. $A \times B = \{(x, y) / x \in A \text{ and } y \in B\}$

similarly, $A \times B \times C = \{(x, y, z) / x \in A, y \in B \text{ and } z \in C\}$

where (x, y, z) is an ordered triple.

Note: If A has m elements and B has n elements, then $A \times B$ has mn elements.

PROPERTIES

- * Commutative property is not true i.e. $A \times B \neq B \times A$
- * Associative property is satisfied i.e. $A \times (B \times C) = (A \times B) \times C$

For example, let $A = \{1, 2\}$, $B = \{3, 4\}$ and $C = \{5\}$

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$(A \times B) \times C = \{(1, 3, 5), (1, 4, 5), (2, 3, 5), (2, 4, 5)\}$$

$$B \times C = \{(3, 5), (4, 5)\}$$

$$A \times (B \times C) = \{(1, 3, 5), (1, 4, 5), (2, 3, 5), (2, 4, 5)\}$$

$$\therefore (A \times B) \times C = A \times (B \times C)$$

- * (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
- (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Proof:

$$\begin{aligned} \text{(i) } A \times (B \cup C) &= \{(x, y) / x \in A \text{ and } y \in B \cup C\} \\ &= \{(x, y) / x \in A \text{ and } (y \in B \text{ or } y \in C)\} \\ &= \{(x, y) / (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C)\} \\ &= \{(x, y) / (x, y) \in A \times B \text{ or } (x, y) \in A \times C\} \\ &= \{(x, y) / (x, y) \in (A \times B) \cup (A \times C)\} \\ &= (A \times B) \cup (A \times C) \end{aligned}$$

similar proof for (ii)

PROBLEMS

- 1) If $A = \{\alpha, \beta\}$ and $B = \{1, 2, 3\}$ what are $A \times B$, $B \times A$, A^2 , B^2 and $(A \times B) \cap (B \times A)$? Is commutative property satisfied?

Solution

$$A \times B = \{(\alpha, 1), (\alpha, 2), (\alpha, 3), (\beta, 1), (\beta, 2), (\beta, 3)\}$$

$$B \times A = \{(1, \alpha), (1, \beta), (2, \alpha), (2, \beta), (3, \alpha), (3, \beta)\}$$

$A \times B \neq B \times A$. $\therefore$ commutative property is not satisfied

$$A^2 = A \times A = \{(\alpha, \alpha), (\alpha, \beta), (\beta, \alpha), (\beta, \beta)\}$$

$$B^2 = B \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

$$(A \times B) \cap (B \times A) = \phi$$

2) If $A = \phi$ and $B = \{1, 2, 3\}$ what are $A \times B$ and $B \times A$?

Solution

$$A \times B = \phi \text{ and } B \times A = \phi$$

RELATIONS

DEFINITION

Let A and B be sets. A binary relation or simply relation from A to B is a subset of $A \times B$. If R is a relation from A to B , then R is a set of ordered pairs where each first element belongs to A and each second element belongs to B .

$(a, b) \in R \Rightarrow$ "a is related to b" written as aRb

$(a, b) \notin R \Rightarrow$ "a is not related to b" written as $a \not R b$.

Eg.

(1) Let $A = \{4, 6, 16\}$ and $B = \{1, 4, 5, 7\}$. Let R be the relation ' $\leq$ ' from A to B . Then $R = \{(4, 4), (4, 5), (4, 7), (6, 7)\}$. Here $(a, b) \in R$ only when $a \leq b$.

(2) Let $A = \{\text{eggs, milk, corn}\}$ and $B = \{\text{cows, goats, hens}\}$. Define a relation R from A to B by $(a, b) \in R$ if 'a' is produced by 'b'.

Then $R = \{(\text{eggs, hens}), (\text{milk, cows}), (\text{milk, goats})\}$

RELATION ON A SET

We say that R is a relation on A if R is a relation from A to itself.

Eg.

Let $A = \{2, 3, 5, 6\}$. Define a relation R from A to A by $(a, b) \in R$ iff 'a divides b'.

Then $R = \{(2, 2), (2, 6), (3, 3), (3, 6), (5, 5), (6, 6)\}$.

Note: We say 'a divides b' written as $a|b$ if $\frac{b}{a}$ gives remainder zero. Eg. $3|27$, $5|20$, $8|64$

TYPES OF RELATIONS

* REFLEXIVE RELATION

A relation R on a set A is said to be reflexive if $(a, a) \in R$ for every $a \in A$. i.e. aRa for every $a \in A$.

Eg.1

Consider the following 5 relations on the set $A = \{1, 2, 3, 4\}$

$$R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(1, 3), (2, 1)\}$$

$$R_4 = \phi \text{ (empty relation)}$$

$$R_5 = A \times A \text{ (universal relation)}$$

Eg.2

- (i) Relation ' $\leq$ ' on the set $\mathbb{Z}$ of integers
- (ii) Relation set inclusion ' $\subseteq$ ' on a collection C of sets
- (iii) Relation ' $\perp$ ' on a set L of all lines in a plane
- (iv) Relation ' $\parallel$ ' on a set L of all lines in a plane
- (v) Relation ' $\mid$ ' on the set $\mathbb{Z}_+$ of all positive integers

In Eg.1, since A contains 1, 2, 3 and 4, a relation R on A is reflexive only if it contains all the 4 pairs $(1, 1), (2, 2), (3, 3), (4, 4)$. $\therefore$ only R_2 and R_5 are reflexive.

In Eg.2, only relation (iii) is not reflexive

since no line is perpendicular to itself.

(i) $a \leq a \quad \forall a \in \mathbb{Z}$ (ii) $A \subseteq A \quad \forall A \in C$ (iv) $a \parallel a \quad \forall a \in L$ (v) $a \mid a \quad \forall a \in \mathbb{Z}_+$

* IRREFLEXIVE RELATION

A relation R on a set A is called irreflexive

if $(a, a) \notin R$ for every $a \in A$. i.e. $a \not R a \quad \forall a \in A$.

In Eg.1, R_3 and R_4 are irreflexive while the rest are not.

In Eg.2, relation (iii) is irreflexive since any line is not

perpendicular to itself, while the rest are not reflexive.

* SYMMETRIC RELATION

A relation R on a set A is symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$ i.e. whenever aRb then bRa , where $a,b \in A$. Thus R is not symmetric if we can find even one $(c,d) \in R$ for which $(d,c) \notin R$, where $c,d \in A$.

In Eg. 1, R_2, R_4 and R_5 are symmetric while R_1 and R_3 are not since $(1,2) \in R_1$ and $(2,1) \notin R_1$ and $(1,3) \in R_3$ and $(3,1) \notin R_3$.

In Eg. 2, relations (iii) and (iv) are symmetric since whenever $a \parallel b$ then $b \parallel a$ and whenever $a \perp b$ then $b \perp a$. Other relations are not symmetric since for eg. (i) $3 \leq 4$ but $4 \not\leq 3$ (ii) $\{1,2\} \subseteq \{1,2,3\}$ but $\{1,2,3\} \not\subseteq \{1,2\}$ (v) $2 \mid 6$ but $6 \nmid 2$.

In the set of all men, the relation 'is the brother of' is symmetric since whenever a is the brother of b , b is the brother of a . In the same set, the relation 'is the father of' is not symmetric, since whenever a is the father of b , then b is not the father of a .

* ANTISYMMETRIC RELATION

A relation R on a set A is antisymmetric if whenever $(a,b) \in R$ and $(b,a) \in R$ then $a=b$ i.e. whenever aRb and bRa then $a=b$. Thus R is not antisymmetric if we can find even one $(c,d) \in R$ and $(d,c) \in R$ for which $c \neq d$.

Another Definition

A relation R on a set A is antisymmetric if whenever $(a,b) \in R$ then $(b,a) \notin R$ where $a \neq b$.

In Eg. 1, R_2 is not antisymmetric since $(1,2) \in R_2$ and $(2,1) \in R_2$ but $1 \neq 2$. Similarly R_5 is not antisymmetric while R_1, R_3 and R_4 are.

In Eg. 2, relation (i) is antisymmetric since $a \leq b$ and $b \leq a \Rightarrow a = b$; ' $\subseteq$ ' is antisymmetric since $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$; Also ' $\parallel$ ' is antisymmetric since $a \parallel b$ and $b \parallel a \Rightarrow a = b$. But ' $\parallel^d$ ' and ' $\perp^r$ ' are not antisymmetric since $a \parallel b$ and $b \parallel a \not\Rightarrow a$ and b are the same. and $a \perp^r b$ and $b \perp^r a \not\Rightarrow a$ and b are the same.

Note that ' $\parallel$ ' on $\mathbb{Z}_+$ is antisymmetric whereas ' $\parallel$ ' on $\mathbb{Z}$ is not antisymmetric since for eg. $3 \parallel -3$ and $-3 \parallel 3$ but $3 \neq -3$.

NOTE:

The properties of being symmetric and antisymmetric are not negatives of each other.

A relation which is neither symmetric and antisymmetric

$$R = \{(1,3), (3,1), (2,3)\} \text{ on } A = \{1,2,3\}$$

A relation which is both symmetric and antisymmetric

$$R = \{(1,1), (2,2)\} \text{ on } A = \{1,2,3\}$$

* TRANSITIVE RELATION

A relation R on a set A is transitive whenever $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$ where $a,b,c \in A$. i.e. whenever aRb and bRc then aRc . Thus R is not transitive if we can find even one $(d,e) \in R$ and $(e,f) \in R$ for which $(d,f) \notin R$.

In Eg. 1, R_3 is not transitive since whenever $(2,1), (1,3) \in R_3$ but $(2,3) \notin R_3$. The rest are all transitive.

In Eg. 2, ' $\leq$ ' is transitive since $a \leq b, b \leq c \Rightarrow a \leq c$, $a, b, c \in \mathbb{Z}$

' $\subseteq$ ' is transitive since $A \subseteq B$ and $B \subseteq C \Rightarrow A \subseteq C, A, B, C \in \mathcal{C}$
 '||' is transitive since $a || b$ and $b || c \Rightarrow a || c, a, b, c \in \mathcal{L}$
 'I' is transitive since $a | b$ and $b | c \Rightarrow a | c, a, b, c \in \mathbb{Z}_+$
 But 'I' is not transitive since $a \nmid b$ and $b \nmid c \nRightarrow a \nmid c, a, b, c \in \mathbb{L}$

In the set of all women, the relation 'is the sister of' is transitive since whenever a is the sister of b and b is the sister of c , then a is a sister of c .
 In the same set, the relation 'is the mother of' is not transitive since if a is the mother of b and b is the mother of c , then a is not the mother of c .

* EQUIVALENCE RELATION

Let A be a nonempty set. A relation R on A is said to be an equivalence relation if R is reflexive, symmetric and transitive.

Eg. Relation ' $||$ ' on the set of all straight lines in a plane.

Counterexample Relation ' $\subseteq$ ' on a collection of sets. It is reflexive and transitive but not symmetric since $A \subseteq B \nRightarrow B \subseteq A$.

* PARTIAL ORDERING RELATION

A relation R on a set A is called a partial ordering relation or a partial order relation if R is reflexive, antisymmetric and transitive. The set A together with the partial ordering R is called a partially ordered set or POSET.

Eg. (1) Relation ' $\geq$ ' on $\mathbb{R}$, the set of all real numbers
 (2) Relation 'divides' on $\mathbb{Z}_+$, the set of all positive integers

Counterexample Relation 'divides' on $\mathbb{Z}$, the set of all integers. It is not antisymmetric since for eg. $5 | -5$ and $-5 | 5$ but $5 \neq -5$

PROBLEMS

1) Consider the following relations on the set $A = \{1, 2, 3\}$

$$R = \{(1, 1), (1, 2), (1, 3), (3, 3)\}$$

$$S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$$

$$T = \{(1, 1), (1, 2), (2, 2), (2, 3)\}$$

$$U = \{(1, 1), (2, 2), (2, 3), (3, 2), (1, 2)\}$$

Determine whether or not each of the above relations on A is reflexive, symmetric, transitive, antisymmetric

Soln.

R is not reflexive, not symmetric, antisymmetric & transitive

S is reflexive, symmetric, not antisymmetric & transitive

T is not reflexive, not symmetric, antisymmetric, not transitive

U is not reflexive, not symmetric, not antisymmetric, not transitive

2) Give examples of relations R on $A = \{1, 2, 3, 4\}$ having the stated property

(i) R is reflexive, symmetric and transitive

(ii) R is reflexive, symmetric but not transitive

(iii) R is symmetric, transitive but not reflexive

Soln.

$$(i) R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$$

$$(ii) R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 3), (2, 1), (3, 2)\}$$

$$(iii) R = \{(1, 1), (3, 3), (1, 3), (3, 1)\}$$

3) Let $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(i, j) / |i - j| = 2\}$

Examine whether R is reflexive, symmetric, antisymmetric, transitive.

Soln.

$$R = \{(1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4)\}$$

R is only symmetric. It is also irreflexive since

$$(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) \notin R.$$

R is not antisymmetric since for eg. $(1, 3) \in R$ and $(3, 1) \in R$ but $1 \neq 3$.

R is not transitive since for eg. $(2, 4) \in R$ and $(4, 2) \in R$ but $(2, 2) \notin R$

- 4) Let $A = \{1, 2, 3, 6\}$. Examine whether 'divides' is a partial ordering relation on A .

Soln.

$$R = \{(1,1), (1,2), (1,3), (1,6), (2,2), (2,6), (3,3), (3,6), (6,6)\}$$

Reflexive

Since $(1,1), (2,2), (3,3), (6,6) \in R$, $(a,a) \in R$ for every $a \in A$.

Hence R is reflexive

Antisymmetric

We see that whenever $(a,b) \in R$ then $(b,a) \notin R$, where $a \neq b$

For eg., $(1,3) \in R$ but $(3,1) \notin R$

$(2,6) \in R$ but $(6,2) \notin R$. Hence R is antisymmetric

Transitive

We see that whenever $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

For eg., $(1,3) \in R$ and $(3,6) \in R \Rightarrow (1,6) \in R$

$(1,2) \in R$ and $(2,6) \in R \Rightarrow (1,6) \in R$.

Hence R is transitive

$\therefore R$ is a partial ordering relation

- 5) Let $S = \{1, 2, 3, \dots, 10\}$ and r be a relation on S defined by $r = \{(x,y) : x+y=10\}$. Is r reflexive, symmetric, transitive?

Soln.

$$r = \{(1,9), (9,1), (2,8), (8,2), (3,7), (7,3), (4,6), (6,4), (5,5)\}$$

r is irreflexive since $(1,1), (2,2), \dots, (10,10) \notin r$

r is symmetric since whenever $(a,b) \in r$, then $(b,a) \in r$

For eg., $(7,3) \in r \Rightarrow (3,7) \in r$

r is not transitive since for eg. $(2,8) \in r$ and $(8,2) \in r \not\Rightarrow (2,2) \in r$

- b) Let $x \sim y$ iff $|x-y| \leq 3$ where $\sim$ is defined on the set of all real numbers R . Is the above relation an equivalence relation?

Soln.

Reflexive

For any $a \in R$, consider $|a-a| = 0 < 3$

$\therefore (a,a) \in \sim$ for any $a \in R$. Hence $\sim$ is reflexive

Symmetric

To check whether whenever $(a, b) \in R$ then $(b, a) \in R$ where $a, b \in \mathbb{R}$.

$$\text{Assume } (a, b) \in R \Rightarrow |a - b| \leq 3$$

$$\Rightarrow |-(a - b)| \leq 3$$

$$\Rightarrow |b - a| \leq 3$$

$$\Rightarrow (b, a) \in R$$

$\therefore R$ is symmetric

Transitive

To check whether whenever $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$, where $a, b, c \in \mathbb{R}$

$$\text{Assume } (a, b) \in R \Rightarrow |a - b| \leq 3 \text{ --- ①}$$

$$\text{Assume } (b, c) \in R \Rightarrow |b - c| \leq 3 \text{ --- ②}$$

$$\begin{aligned} \text{Consider } |a - c| &= |\overline{a - b} + \overline{b - c}| \\ &\leq |a - b| + |b - c| \text{ (from ① and ②)} \\ &\leq 3 + 3 = 6 \end{aligned}$$

$$|a - c| \leq 6$$

Hence $(a, c) \notin R$

$\therefore R$ is not transitive and hence not an equivalence relation.

DEFINITION

Congruence Modulo m

x is said to be congruent to y modulo m if $x - y$ is divisible by m , i.e. if $x - y = km$ for some integer k . This is denoted by $x \equiv y \pmod{m}$.

7) Prove that the relation "congruence modulo m " given by $R = \{(x, y) / x - y \text{ is divisible by } m\}$ over the set of integers is an equivalence relation.

Soln.

Reflexive

For any $x \in \mathbb{Z}$, consider $x - x = 0$
 $= 0 \times m$
 $\Rightarrow x - x$ is divisible by m

$\Rightarrow (x, x) \in R$ for any $x \in \mathbb{Z}$
 $\therefore R$ is reflexive

Symmetric

For any $x, y \in \mathbb{Z}$, to prove that whenever $(x, y) \in R$ then $(y, x) \in R$.

Assume $(x, y) \in R \Rightarrow x - y$ is divisible by m | Eg.
 $\Rightarrow -(x - y)$ is divisible by m | 10 is divisible by 5
 $\Rightarrow y - x$ is divisible by m | $\Rightarrow -10$ " " " 5
 $\Rightarrow (y, x) \in R$

$\therefore R$ is symmetric

Transitive

For any $x, y, z \in \mathbb{Z}$, to prove that whenever $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$.

Assume $(x, y) \in R \Rightarrow x - y$ is divisible by m | Eg.
 Assume $(y, z) \in R \Rightarrow y - z$ is divisible by m | 9 is divisible by 3
 $\Rightarrow 9 + 27 = 36$ " " " 3

Then the sum $x - y + y - z$ is divisible by m
 $\Rightarrow x - z$ is divisible by m
 $\Rightarrow (x, z) \in R$

$\therefore R$ is transitive

Hence R is an equivalence relation

8) Let $X = \{1, 2, 3, \dots, 7\}$ and let $R = \{(x, y) / x - y \text{ is divisible by } 3\}$ be a relation on X . show that R is an equivalence relation.

Soln.

$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (1, 4), (1, 7), (2, 5), (3, 6), (4, 7), (4, 1), (5, 2), (6, 3), (7, 1), (7, 4)\}$

Reflexive

Since $(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7) \in R$,

we see that $(a, a) \in R \forall a \in X$

$\therefore R$ is reflexive

Symmetric

For any $a, b \in X$, we see that whenever $(a, b) \in R$ then $(b, a) \in R$
 For eg., $(1, 7) \in R \Rightarrow (7, 1) \in R$

$$(3,6) \in R \Rightarrow (6,3) \in R$$

$\therefore R$ is symmetric

Transitive

For any $a, b, c \in X$, we see that whenever $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

$$\text{For eg., } (1,7) \in R \text{ and } (7,4) \in R \Rightarrow (1,4) \in R$$

$$(2,2) \in R \text{ and } (2,5) \in R \Rightarrow (2,5) \in R$$

$\therefore R$ is transitive

Hence R is an equivalence relation.

9) Prove that $R = \{(x,y) / x-y \text{ is divisible by } 19\}$ defined on the set of all integers is an equivalence relation
Hint: similar to Problem 7

10) Let $X = \{1, 2, 3, \dots, 20\}$ and $R = \{(x,y) / x-y \text{ is divisible by } 5\}$ be a relation on X . Show that R is an equivalence relation.
Hint: similar to Problem 8

11) Let A be a set of nonzero integers and let $\sim$ be a relation on A defined by $(a,b) \sim (c,d)$ whenever $ad = bc$. Prove that $\sim$ is an equivalence relation.

Soln.

Reflexive

For any $(a,b) \in A \times A$, we know that $ab = ba$ since $a, b \in A$
 $\Rightarrow (a,b) \sim (a,b)$ for any $(a,b) \in A \times A$

$\therefore \sim$ is reflexive

Symmetric

To prove that whenever $(a,b) \sim (c,d)$ then $(c,d) \sim (a,b)$ for any $(a,b), (c,d) \in A \times A$

$$\text{Assume } (a,b) \sim (c,d) \Rightarrow ad = bc$$

$$\Rightarrow cb = da$$

$$\Rightarrow (c,d) \sim (a,b) \text{ (by definition)}$$

$\therefore \sim$ is symmetric

Transitive

To prove that whenever $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$ then $(a,b) \sim (e,f)$ for any $(a,b), (c,d), (e,f) \in A \times A$.

Assume $(a,b) \sim (c,d) \Rightarrow ad=bc$ — ①

Assume $(c,d) \sim (e,f) \Rightarrow cf=de$ — ②

$$\textcircled{1} \times \textcircled{2} \Rightarrow adcf = bcde$$

$$\Rightarrow af = be$$

$$\Rightarrow (a,b) \sim (e,f) \text{ (by definition)}$$

$\therefore \sim$ is transitive

Hence $\sim$ is an equivalence relation.

12) If R and S are relations on a set A , prove that

(i) If R and S are reflexive, then $R \cup S$ and $R \cap S$ are both reflexive

(ii) If R and S are symmetric, then $R \cup S$ and $R \cap S$ are both symmetric

(iii) If R and S are transitive, then $R \cap S$ is also transitive but $R \cup S$ need not be transitive

Soln.

(i) Since R and S are reflexive,

$(a,a) \in R$ for all $a \in A$ and $(a,a) \in S$ for all $a \in A$

$\Rightarrow (a,a) \in R \cup S$ for all $a \in A$ and $(a,a) \in R \cap S$ for all $a \in A$

$\therefore R \cup S$ and $R \cap S$ are reflexive

(ii) To prove that whenever $(a,b) \in R \cup S$ then $(b,a) \in R \cup S$

Assume $(a,b) \in R \cup S \Rightarrow (a,b) \in R$ or $(a,b) \in S$

$\Rightarrow (b,a) \in R$ or $(b,a) \in S$ ($\because R$ and S are

$\Rightarrow (b,a) \in R \cup S$ symmetric)

$\therefore R \cup S$ is symmetric

Now to prove that whenever $(a,b) \in R \cap S$ then $(b,a) \in R \cap S$

Assume $(a,b) \in R \cap S \Rightarrow (a,b) \in R$ and $(a,b) \in S$

$\Rightarrow (b,a) \in R$ and $(b,a) \in S$

$\Rightarrow (b,a) \in R \cap S$

$\therefore R \cap S$ is symmetric

(iii) To prove that whenever $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$ then $(a,c) \in R \cap S$

Assume $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$

$\Rightarrow [(a,b) \in R \text{ and } (a,b) \in S] \text{ and } [(b,c) \in R \text{ and } (b,c) \in S]$

$\Rightarrow [(a,b) \in R \text{ and } (b,c) \in R] \text{ and } [(a,b) \in S \text{ and } (b,c) \in S]$

$\Rightarrow (a,c) \in R$ and $(a,c) \in S$ ($\because R$ and S are transitive) $\therefore$

$\Rightarrow (a,c) \in R \cap S$

$\therefore R \cap S$ is transitive

To prove that $R \cup S$ need not be transitive, it is enough if we give an example in which R and S are transitive but $R \cup S$ is not.

Let $A = \{1, 2, 3\}$

and let $R = \{(1,2), (2,3), (1,3)\}$ and $S = \{(2,1), (3,2), (3,1)\}$ be relations on A . Note that R and S are transitive.

Now $R \cup S = \{(1,2), (2,3), (1,3), (2,1), (3,2), (3,1)\}$

$(2,3) \in R \cup S$ and $(3,2) \in R \cup S$ but $(2,2) \notin R \cup S$

$\therefore R \cup S$ is not transitive

EQUIVALENCE CLASS

DEFINITION

Let R be an equivalence relation on a set A . For any $a \in A$, the set $[a]_R$ given by

$$[a]_R = \{x \in A / aRx\}$$

is called an R equivalence class generated by $a \in A$.

Note: Obviously $[a]$ is a nonempty subset of A since $a \in [a]$ for aRa ($\because R$ is an equivalence relation)

THEOREM

Prove that every equivalence relation R on a set A generates a unique partition

PROOF:

Before proving the main theorem, let us prove the following two theorems.

→ Eg. for Equivalence class
Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (2,2), (3,3), (4,4), (1,3), (3,1), (1,4), (4,1), (3,4), (4,3)\}$ be an equivalence relation on A . Then $[1]_R = \{1, 3, 4\}$, $[2]_R = \{2\}$
 $[3]_R = \{1, 3, 4\}$, $[4]_R = \{1, 3, 4\}$

THEOREM ①

Let R be an equivalence relation on a set A . Then any two equivalence classes $[a]$ and $[b]$ are equal iff aRb .

PROOF:

Assume $[a] = [b]$. To prove that aRb .

Since $a \in [a]$, $a \in [b]$ ($\because [a] = [b]$)
 $\Rightarrow aRb$

Conversely,

Assume aRb . To prove that $[a] = [b]$

Let $x \in [a] \Rightarrow xRa$

$\Rightarrow xRa$ and aRb (by assumption)

$\Rightarrow xRb$ ($\because R$ is transitive)

$\Rightarrow x \in [b]$

$\therefore [a] \subseteq [b] \text{ --- ①}$

Let $x \in [b] \Rightarrow xRb$

$\Rightarrow xRb$ and aRb (by assumption)

$\Rightarrow xRb$ and bRa ($\because R$ is symmetric)

$\Rightarrow xRa$ ($\because R$ is transitive)

$\Rightarrow x \in [a]$

$\therefore [b] \subseteq [a] \text{ --- ②}$

From ① and ②, $[a] = [b]$

THEOREM ②

Let R be an equivalence relation on a set A and let $a, b \in A$. Then aRb is impossible iff $[a] \cap [b] = \emptyset$ i.e. $a \not R b$ iff their respective classes are disjoint.

PROOF:

Let $a \not R b$. To prove that $[a] \cap [b] = \emptyset$. Let us assume the contrary. Assume $[a] \cap [b] \neq \emptyset$.

$\Rightarrow$ There exists an element x such that

$x \in [a] \cap [b]$

$\Rightarrow x \in [a]$ and $x \in [b]$

$\Rightarrow xRa$ and xRb

$\Rightarrow aRx$ and xRb ($\because R$ is symmetric)

$\Rightarrow aRb$ ($\because R$ is transitive)

which is a contradiction since $a \not R b$.

∴ Our Assumption is wrong

Hence $[a] \cap [b] = \emptyset$

Conversely,

Let $[a] \cap [b] = \emptyset$. To prove that $a \not R b$. Let us assume the contrary, i.e. Assume $a R b$.

$\Rightarrow a \in [b]$ We already know $a \in [a]$

$\Rightarrow a \in [a] \cap [b]$

$\Rightarrow [a] \cap [b] \neq \emptyset$ which is a

contradiction since $[a] \cap [b] = \emptyset$. ∴ our assumption is wrong.

Hence $a \not R b$.

PROOF OF MAIN THEOREM

Let R be an equivalence relation on a set A . Let $[a]$ be the equivalence class generated by the element $a \in A$.

(i) Since R is reflexive, $a R a \Rightarrow a \in [a]$

∴ Each $[a]$ is a nonempty subset of A

(ii) For any two elements $a, b \in A$ either $a R b$ or $a \not R b$

$\Rightarrow [a] = [b]$ or $[a] \cap [b] = \emptyset$ (by Theorems ① and ②)

$\Rightarrow$ The equivalence classes are all mutually disjoint

(iii) Since for any $a \in A$, we know that $a \in [a]$,

we see that union of all the equivalence classes is equal to A .

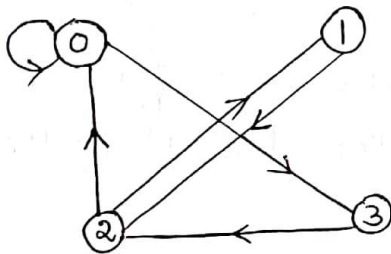
Therefore the set of all equivalence classes generated by the elements of A forms a partition of A .

GRAPHS OF RELATIONS

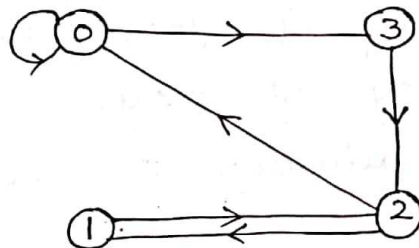
A graph is an illustration which gives a clearer picture of the situation under consideration. The graph of the function $y=5x+7$ helps us to understand how the function behaves and tells us that this function is a straight line.

Let $A=\{0,1,2,3\}$ be a set and let R be the relation $\{(0,0), (0,3), (1,2), (2,1), (3,2), (2,0)\}$ on A . The elements of A are called the vertices of the graph. Place the vertices enclosed in a circle or denoted by a point. If aRb i.e. $(a,b) \in R$ then connect vertex a to vertex b with an arrow directed towards b called an edge of the graph. This type of graph of a relation R is called a directed graph or digraph. Vertex is also called as node, point, junction and edge is also called as arc, line, branch.

The digraph for the above relation is

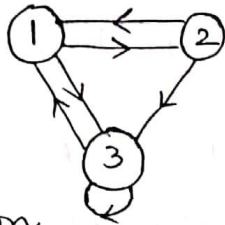


The actual location of the vertices is immaterial. Place the vertices in such a way that the graph looks neat and is easy to read. It does not matter if two graphs of a given relation look different. Thus I can represent the above graph more neatly as follows.



PROBLEMS

- 1) Find out the set A and the relation R on A for the following digraph.



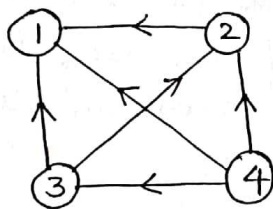
Solution

$$A = \{1, 2, 3\} \text{ and } R = \{(1, 3), (3, 1), (1, 2), (2, 1), (2, 3), (3, 3)\}$$

- 2) Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) / x > y\}$. Draw the graph of R .

Solution

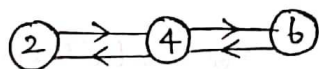
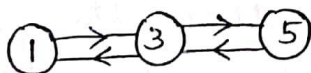
$$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$



- 3) Consider the following relation on $A = \{1, 2, \dots, 6\}$. Let $R = \{(i, j) / |i - j| = 2\}$. Draw the digraph of R .

Solution

$$R = \{(1, 3), (3, 1), (2, 4), (4, 2), (5, 3), (3, 5), (4, 6), (6, 4)\}$$



- 4) The following relations are defined on $\{1, 3, 5\}$
Let r be the relation xry iff $y = x + 2$ and
 s be the relation xsy iff $x < y$.
Find rs , sr and also illustrate rs , sr via a digraph.

Solution

$$A = \{1, 3, 5\}$$

$$r = \{(1, 3), (3, 5)\} \quad s = \{(1, 3), (1, 5), (3, 5)\}$$

$$rs = \{(1, 5)\} \quad sr = \{(1, 5)\}$$

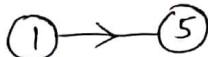
Graph

rs



(3)

sr



(3)

5) Let $A = \{1, 2, 3, 4, 5\}$. Define the relation r on A by xry defined only if $y = x + 1$. Find r , r^2 and r^3 and find their digraphs.

Solution

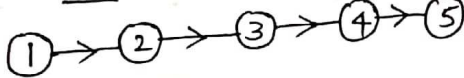
$$r = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$$

$$r^2 = \{(1, 3), (2, 4), (3, 5)\}$$

$$r^3 = \{(1, 4), (2, 5)\}$$

Graph

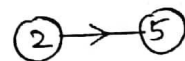
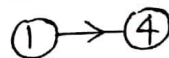
r



r^2



r^3

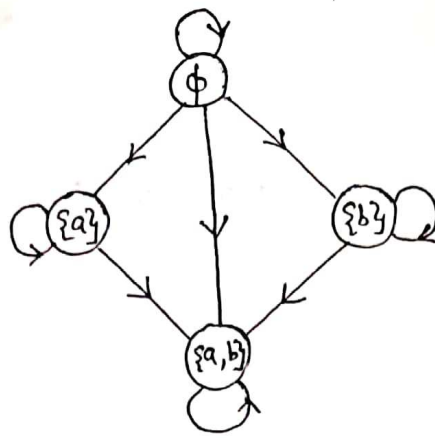


6) Let $B = \{a, b\}$ and let $A = P(B)$. If R is the relation $\subseteq$ on A , find its digraph

Solution

$$A = P(B) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

$$R = \{(\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{a, b\}), (\{a\}, \{a, b\}), (\{b\}, \{a, b\}), (\emptyset, \emptyset), (\{a\}, \{a\}), (\{b\}, \{b\}), (\{a, b\}, \{a, b\})\}$$



7) Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) / x > y\}$. Draw the graph of R .

HASSE DIAGRAM OR ORDERING DIAGRAM

Rules for drawing Hasse Diagram are:

- (1) Each element is related to itself (so an edge from a vertex to itself is not necessary)
- (2) If vertex 'b' appears above vertex 'a' and if vertex 'a' is connected to vertex 'b', then aRb
- (3) If vertex 'c' is above vertex 'a' and if 'c' is connected to 'a' by a sequence of edges, then aRc
- (4) Vertices are denoted by points rather than circles

Remark:

From (1), we see that the relation needs to be reflexive
 from (2), we see that it needs to be antisymmetric
 from (3), we see that it needs to be transitive
 Thus Hasse Diagram can be drawn only for partial ordering relations and hence is also called ordering diagram.

PROBLEMS

1) Let $A = \{1, 2, 3, 4\}$ and let R be the relation ' $\leq$ ' on A .
 Draw the Hasse Diagram of R .

Solution

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

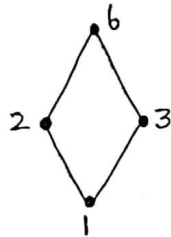
Hasse Diagram



- 2) Let $A = \{1, 2, 3, 6\}$ and let the relation defined on A be 'divides'. Draw the Hasse Diagram.

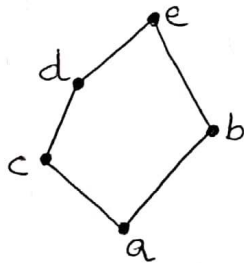
Solution

$$R = \{(1,1), (2,2), (3,3), (6,6), (1,2), (1,3), (1,6), (2,6), (3,6)\}$$



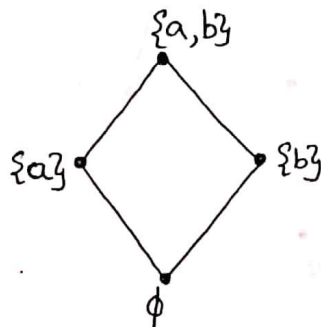
- 3) For the following relation $R = \{(a,a), (b,b), (c,c), (d,d), (e,e), (a,c), (a,d), (a,e), (a,b), (c,d), (c,e), (d,e), (b,e)\}$ on $A = \{a, b, c, d, e\}$, draw Hasse diagram

Solution



- 4) Let $B = \{a, b\}$ and $A = P(B)$. Draw Hasse diagram for the relation ' $\subseteq$ ' on A.

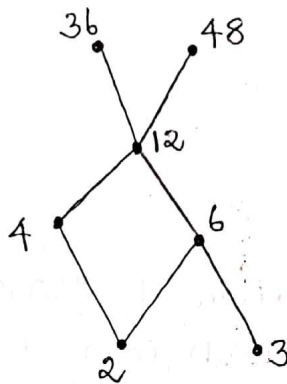
Solution



- 5) Let S be the relation ' $\mid$ ' on B where $B = \{2, 3, 4, 6, 12, 36, 48\}$. Draw the Hasse Diagram for the relation S .

Solution

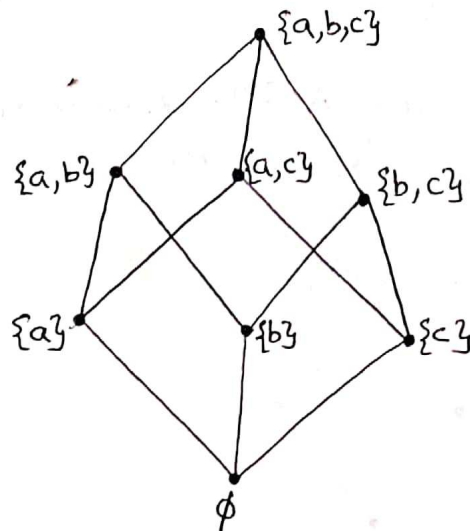
$$S = \{(2, 2), (2, 4), (2, 6), (2, 12), (2, 36), (2, 48), (3, 3), (3, 6), (3, 12), (3, 36), (3, 48), (4, 4), (4, 12), (4, 36), (4, 48), (6, 6), (6, 12), (6, 36), (6, 48), (12, 12), (12, 36), (12, 48), (36, 36), (48, 48)\}$$



- 6) Let $B = \{a, b, c\}$ and $P(B) = A$ be its power set. Draw Hasse diagram for the relation $\subseteq$ on A .

Solution

$$A = P(B) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$



GROUP THEORYBINARY OPERATION

Let A be a nonempty set. A mapping $f: A \times A \rightarrow A$ is called a binary operation. We normally denote binary operation using $*$ or $\cdot$.

PROPERTIES OF BINARY OPERATIONS

Let $*$: $G \times G \rightarrow G$ be a binary operation where G is a nonempty set.

① CLOSURE PROPERTY

$*$ is closed if for any $a, b \in G$, then $a * b \in G$.

② ASSOCIATIVE PROPERTY

$a * (b * c) = (a * b) * c$ for any $a, b, c \in G$.

③ EXISTENCE OF IDENTITY ELEMENT

There exists an element $e \in G$ such that

$$a * e = e * a = a \quad \forall a \in G$$

The element ' e ' is called the identity element of G .

④ EXISTENCE OF INVERSE

For any $a \in G$, if there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$, then a^{-1} is called the inverse of a .

⑤ COMMUTATIVE OR ABELIAN PROPERTY

$a * b = b * a$ for any $a, b \in G$

⑥ DISTRIBUTIVE PROPERTY

$a * (b * c) = (a * b) * c$ Left Distributive law

$(b * c) * a = (b * a) * c$ Right Distributive law

for any $a, b, c \in G$

⑦ CANCELLATION PROPERTY

$a * b = a * c \Rightarrow b = c$ Left Cancellation law

$b * a = c * a \Rightarrow b = c$ Right Cancellation law

for any $a, b, c \in G$

⑧ IDEMPOTENT PROPERTY

$a * a = a \quad \forall a \in G$.

EXAMPLES

1. The operation '-' on $\mathbb{N}$ is not closed since $2, 3 \in \mathbb{N}$ but $2-3 = -1 \notin \mathbb{N}$
2. The operation $*$ defined by $a*b = \frac{a+b}{2}$ is closed in $\mathbb{R}$ but not in $\mathbb{Z}$.

3. Addition and multiplication are commutative on $\mathbb{Z}$ whereas subtraction is not.

4. Matrix multiplication on the set of all 2×2 matrices is not commutative but is associative.

For eg., $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 15 & 10 \end{pmatrix}$

but $\begin{pmatrix} 5 & 6 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 23 & 24 \\ -6 & -8 \end{pmatrix} \therefore$ Not commutative

5. For $\mathbb{Z}$, '0' is the identity element under addition and '1' is the identity element under multiplication.

6. For any $a \in \mathbb{Z}$, $-a$ is the inverse of a under addition since $a+(-a) = (-a)+a = 0$

7. $+$ and $-$ in $\mathbb{Z}$ do satisfy both the cancellation laws. But matrix multiplication does not satisfy cancellation law

For eg., let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ $C = \begin{pmatrix} 0 & -3 \\ 1 & 5 \end{pmatrix}$

Then $AB = AC = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$

But $B \neq C$

PROBLEMS

1) Let $*$ be a binary operation defined on $\mathbb{R} - \{1\}$ defined by $x*y = x+y-xy$. Show that $*$ is commutative and associative. Find the identity element and indicate inverse of each element.

Solution

(i) To P.T. $a*b = b*a$ for any $a, b \in \mathbb{R} - \{1\}$

$$\begin{aligned} \text{LHS} &= a*b \\ &= a+b-ab \\ &= b+a-ba \\ &= b*a \\ &= \text{RHS.} \end{aligned}$$

$\therefore *$ is commutative

(ii) To P.T. $a * (b * c) = (a * b) * c$ for any $a, b, c \in R - \{1\}$ (2)

$$\begin{aligned} \text{LHS} &= a * (b * c) = a * (b + c - bc) = a + b + c - bc - a(b + c - bc) \\ &= a + b + c - ab - bc - ca + abc \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (a * b) * c = (a + b - ab) * c = a + b - ab + c - (a + b - ab)c \\ &= a + b + c - ab - bc - ca + abc \end{aligned}$$

$$\text{LHS} = \text{RHS} \therefore * \text{ is associative}$$

(iii) To P.T. $\exists$ an element $e \in R - \{1\}$ such that $a * e = a$ for any $a \in R - \{1\}$.

Let $a \in R - \{1\}$

$$\text{Consider } a * e = a$$

$$\Rightarrow a + e - ae = a$$

$$\Rightarrow e(1 - a) = 0$$

$$\Rightarrow e = 0 \quad (\because a \neq 1 \text{ as } a \in R - \{1\})$$

$$e \in R - \{1\}$$

$\therefore 0 \in R - \{1\}$ is the identity element

(iv) To P.T. for $a \in R - \{1\}$, $\exists$ an element $a^{-1} \in R - \{1\}$ such that

$$a * a^{-1} = a^{-1} * a = e$$

Let $a \in R - \{1\}$

$$\text{Consider } a * a^{-1} = e$$

$$\Rightarrow a + a^{-1} - aa^{-1} = e$$

$$\Rightarrow a + a^{-1}(1 - a) = 0$$

$$\Rightarrow a^{-1} = \frac{-a}{1-a} \in R - \{1\} \quad \left[\begin{array}{l} \text{Reason} \\ \text{If } \frac{-a}{1-a} = 1, \text{ then } -a = 1 - a \\ \Rightarrow 0 = 1 \end{array} \right]$$

which is impossible

$\therefore$ Inverse element for

$$a \text{ is } \frac{-a}{1-a}$$

$$\therefore \frac{-a}{1-a} \neq 1$$

2) Let $*$ denote a binary operation on N given by $x * y = x$. Show that $*$ is not commutative but is associative. Which elements are idempotent?

Solution

(i) Let $a, b \in N$.

$$a * b = a \text{ and } b * a = b$$

$\therefore a * b \neq b * a$ Hence $*$ is not commutative.

(ii) To P.T. $a*(b*c) = (a*b)*c$ for any $a, b, c \in \mathbb{N}$

$$\text{LHS} = a*(b*c) = a*b = a$$

$$\text{RHS} = (a*b)*c = a*c = a$$

$\therefore \text{LHS} = \text{RHS}$ Hence $*$ is associative

(iii) For any $a \in \mathbb{N}$, $a*a = a$ (by definition)

$\therefore$ All the elements are idempotent.

3) Let $x*y = \text{lcm}(x, y)$ where $*$ is a binary operation on I the set of positive integers. S.T. $*$ is commutative and associative. Find the identity element and also state which elements are idempotent.

Solution

(i) Let $a, b \in I$

$$a*b = \text{lcm}(a, b) = \text{lcm}(b, a) = b*a$$

$$\therefore a*b = b*a \text{ for any } a, b \in I$$

$\therefore *$ is commutative

(ii) Let $a, b, c \in I$

$$a*(b*c) = a*\text{lcm}(b, c) = \text{lcm}(a, \text{lcm}(b, c))$$

$$(a*b)*c = \text{lcm}(a, b)*c = \text{lcm}(\text{lcm}(a, b), c)$$

Note that $a*(b*c) = (a*b)*c$ since lcm of 3 integers taken in any order will be the same.

(iii) For any $a \in I$, to prove that $\exists$ an element $e \in I$ such that $a*e = e*a = a$

$$\text{Consider } a*e = a$$

$$\Rightarrow \text{lcm}(a, e) = a$$

$$\Rightarrow e = 1 \in I \text{ is the identity element.}$$

(iv) $a*a = \text{lcm}(a, a) = a$ for any $a \in I$

$\therefore$ Each element in I is idempotent.

4) Prove that the set $\mathbb{N}$ satisfies associative and identity properties under the operation $x*y = \max(x, y)$

Solution

(i) Let $a, b, c \in \mathbb{N}$.

$$a * (b * c) = a * \max(b, c) = \max(a, \max(b, c))$$

$$(a * b) * c = \max(a, b) * c = \max(\max(a, b), c)$$

$\therefore a * (b * c) = (a * b) * c$ since maximum of 3 natural numbers taken in any order will be the same.

(ii) For any $a \in \mathbb{N}$, to P.T. $\exists$ an element $e \in \mathbb{N}$ such that $a * e = e * a = a$.

Let $a \in \mathbb{N}$. Consider $a * e = a$

$$\Rightarrow \max(a, e) = a$$

$\Rightarrow e = 1 \in \mathbb{N}$ is the identity element.

DEFINITION — SEMIGROUP

Let $*$ be a binary operation defined on a nonempty set S . Then S is said to be a semigroup if $*$ satisfies CLOSURE & ASSOCIATIVE PROPERTIES. We denote the semigroup by $(S, *)$

DEFINITION — MONOID

Let $*$ be a binary operation defined on a nonempty set S . Then S is said to be a monoid if $*$ satisfies CLOSURE, ASSOCIATIVE PROPERTY and IDENTITY ELEMENT exists.
i.e. A semigroup with identity property is a monoid.

EXAMPLES

1. $(\mathbb{N}, \times)$ is a monoid with identity element '1' but $(\mathbb{N}, +)$ is not a monoid since the additive identity '0' is not a natural number.
2. Let $E = \{2, 4, 6, 8, \dots\}$. Then $(E, +)$ and $(E, \times)$ are semigroups but not monoids.
3. Let S be a nonempty set and $P(S)$ be its power set. Then $(P(S), \cup)$ and $(P(S), \cap)$ are monoids with identity elements ϕ and S respectively.
4. Let $\mathbb{Z}_n$ be the set of all congruence classes modulo n . Define $+_n$ by $[a] +_n [b] = (a+b) \bmod n$, where $[a], [b] \in \mathbb{Z}_n$ and $\times_n$ by $[a] \times_n [b] = (ab) \bmod n$. Then $(\mathbb{Z}_n, +_n)$ and $(\mathbb{Z}_n, \times_n)$ are monoids.

For eg., when $n=5$ consider the Cayley's table for t_5 and x_5 . (For convenience, we denote $[a]$ by a itself.)

t_5	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

x_5	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

We see that associative property is satisfied and $[0]$ and $[1]$ are the identities w.r.t t_n and x_n respectively.

DEFINITION - ABELIAN MONOID

A monoid $(S, *)$ is said to be abelian if it satisfies commutative property i.e. if $a * b = b * a \forall a, b \in S$

Eg.

$\mathbb{Z}$, $\mathbb{R}$ and $\mathbb{C}$ are abelian monoids under usual addition and multiplication.

DEFINITION - CYCLIC MONOID

A monoid $(S, *)$ is said to be cyclic if there exists an element $a \in S$ such that for any $x \in S$, $x = a^n$ (Here $a^n = a * a * a \dots n$ times) where n is an integer. Then S is said to be generated by the element 'a' and 'a' is called the GENERATOR of S .

Eg.

① Let $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$ be the set of all non -ve integers. Then $(\mathbb{Z}_+, +)$ is an infinite cyclic monoid since $1 \in \mathbb{Z}_+$ is the generator of $\mathbb{Z}_+$.

$1^0 = 0$ (write 1 no number of times)

$1^1 = 1, 1^2 = 1+1=2, 1^3 = 1+1+1=3 \dots$

② $G = \{1, -1, i, -i\}$ is a cyclic monoid under multiplication. Here i is a generator since $1 = i^4, -1 = i^2, i = i^1$ and $-i = i^3$

$$a * (b * c) = a * \max(b, c) = \max(a, \max(b, c))$$

$$(a * b) * c = \max(a, b) * c = \max(\max(a, b), c)$$

$\therefore a * (b * c) = (a * b) * c$ since maximum of 3 natural numbers taken in any order will be the same.

(ii) For any $a \in \mathbb{N}$, to P.T. $\exists$ an element $e \in \mathbb{N}$ such that $a * e = e * a = a$.

Let $a \in \mathbb{N}$. Consider $a * e = a$

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$$a * (b * c) = a * \max(b, c) = \max(a, \max(b, c))$$

$$(a * b) * c = \max(a, b) * c = \max(\max(a, b), c)$$

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(ii) For any $a \in \mathbb{N}$, to P.T. $\exists$ an element $e \in \mathbb{N}$ such that $a * e = e * a = a$.

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Note, that $-i$ is also a generator of $(G, \times)$, since $1 = (-i)^4$, $-1 = (-i)^2$, $i = (-i)^3$ and $-i = (-i)^1$. (4)

GROUP

DEFINITION - GROUP

Let G be a nonempty set with binary operation $*$. Then $(G, *)$ will be called a group if the following properties are satisfied.

(i) CLOSURE PROPERTY

For any $a, b \in G$, $a * b \in G$.

(ii) ASSOCIATIVE PROPERTY

For any $a, b, c \in G$, $a * (b * c) = (a * b) * c$

(iii) EXISTENCE OF IDENTITY

For any $a \in G$, there exists an element $e \in G$ such that $a * e = e * a = a$

(iv) EXISTENCE OF INVERSE

For any $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

Here $(G, *)$ represents a group G with $*$ as the binary operation.

DEFINITION - ABELIAN GROUP

A group $(G, *)$ is said to be abelian if it satisfies commutative property, i.e. $a * b = b * a \forall a, b \in G$.

EXAMPLES

1. $(\mathbb{Z}, +)$ is a group but $(\mathbb{Z}, \times)$ is not a group since inverse does not exist (For eg. $2 \in \mathbb{Z}$ but $\frac{1}{2} \notin \mathbb{Z}$)
2. $(\mathbb{I}, +)$ and $(\mathbb{R}, +)$ are abelian groups.
3. The set of all 3×3 matrices is an abelian group under addition. It is also a group under multiplication but is not abelian, since matrix multiplication is not commutative.

DEFINITION - ORDER OF A GROUP

The order of a group $(G, *)$ denoted by $O(G)$ or $|G|$ is the number of elements in G .

If $O(G)$ is a finite number, then the group G is called a finite group. Otherwise it is called an infinite group.

Eg.

$G = \{1, -1, i, -i\}$ under multiplication is a finite group of order 4 whereas $(\mathbb{Z}, +)$ is an infinite abelian group.

PROBLEMS

- 1) Prove that $G = \{1, -1, i, -i\}$ is a group under usual multiplication.

Solution

<u>Cayley's Table</u>				
$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

From the table, we find the operation is closed and is associative. Identity element is 1.

Inverse of 1 is 1

Inverse of -1 is -1

Inverse of i is -i

Inverse of -i is i

$\therefore G$ forms a group under multiplication

- 2) Prove that $A = \{1, \omega, \omega^2\}$ is an abelian group under multiplication where $1, \omega, \omega^2$ are cube roots of unity and $\omega^3 = 1$.

Solution

<u>Cayley's Table</u>			
$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

From Cayley's table, we find that $*$ is closed, associative and commutative. Identity element is 1.

Inverse of 1 is 1

Inverse of ω is ω^2

Inverse of ω^2 is ω

$\therefore (A, *)$ forms an abelian group.

- 3) Write down the composition table for $(\mathbb{Z}_3, +_3)$ and prove that it is an abelian group.

(5)

Solution

$$Z_3 = \{0, 1, 2\}$$

Composition Table

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

From the table, we see that $+_3$ satisfies closure, associative and commutative properties. Identity is 0.

Inverse of 0 is 0

Inverse of 1 is 2

Inverse of 2 is 1

$\therefore (Z_3, +_3)$ is an abelian group.

4) Prove that the set of matrices $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ forms a group under multiplication.

Solution

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and let $G = \{A, B\}$

To prove that $(G, \times)$ is a group.

$$\text{Now } A \times A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A$$

Similarly we find that $AB = B$, $BA = B$ and $BB = A$.

$\times$	A	B
A	A	B
B	B	A

From the table, we find that $\times$ is closed, associative. Identity element is $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
Inverse of A is A and inverse of B is B.
 $\therefore (G, \times)$ is a group.

5) Prove that the set of all non zero real numbers forms an infinite abelian group under the operation $*$ defined by $a * b = \frac{ab}{2} \forall a, b \in \mathbb{R} - \{0\}$

Solution① CLOSURE PROPERTY

Let $a, b \in \mathbb{R} - \{0\}$

$$a * b = \frac{ab}{2} \in \mathbb{R} - \{0\} \text{ since } a \neq 0 \text{ and } b \neq 0$$

$\therefore *$ satisfies closure property.

② ASSOCIATIVE PROPERTY

Let $a, b, c \in \mathbb{R} - \{0\}$

To P.T. $a * (b * c) = (a * b) * c$.

$$a * (b * c) = a * \frac{bc}{2} = \frac{abc}{2 \times 2} = \frac{abc}{4}$$

$$(a*b)*c = \left(\frac{ab}{2}\right)*c = \frac{abc}{2 \times 2} = \frac{abc}{4}$$

$$\therefore a*(b*c) = (a*b)*c$$

③ IDENTITY ELEMENT

For any $a \in R - \{0\}$, to prove that there exists an element $e \in R - \{0\}$ such that $a*e = e*a = a$.

Let $a \in R - \{0\}$

Consider $a*e = a$

$$\Rightarrow \frac{ae}{2} = a$$

$$\Rightarrow e = 2 \in R - \{0\}$$

$\therefore 2$ is the identity element

④ INVERSE ELEMENT

For any $a \in R - \{0\}$, to prove that there exists an element $a^{-1} \in R - \{0\}$ such that $a*a^{-1} = a^{-1}*a = e$

Let $a \in R - \{0\}$

Consider $a*a^{-1} = e$

$$\Rightarrow \frac{aa^{-1}}{2} = 2$$

$$\Rightarrow a^{-1} = \frac{4}{a} \in R - \{0\}$$

$\therefore a^{-1} = \frac{4}{a}$ is the inverse of a . $\therefore$ Inverse exists.

⑤ COMMUTATIVE PROPERTY

For any $a, b \in R - \{0\}$, to prove that $a*b = b*a$

$$a*b = \frac{ab}{2} = \frac{ba}{2} = b*a$$

$\therefore R - \{0\}$ forms an abelian group.

6) In $\mathbb{Z}$, define $*$ by $a*b = a+b+1$. Show that $*$ satisfies all the properties of a group.

Solution

① CLOSURE PROPERTY

Let $a, b \in \mathbb{Z}$

$a*b = a+b+1 \in \mathbb{Z}$ ($\because a, b$ are integers, $a+b+1$ will also be an integer)

$\therefore *$ satisfies closure property.

② ASSOCIATIVE PROPERTY

Let $a, b, c \in \mathbb{Z}$

$$a * (b * c) = a * (b + c + 1) = a + b + c + 1 + 1 = a + b + c + 2$$

$$(a * b) * c = (a + b + 1) * c = a + b + 1 + c + 1 = a + b + c + 2$$

③ IDENTITY ELEMENT

To find an element $e \in \mathbb{Z}$ such that $a * e = e * a = a \forall a \in \mathbb{Z}$

Let $a \in \mathbb{Z}$. Consider $a * e = a$

$$\Rightarrow a + e + 1 = a$$

$$\Rightarrow e = -1 \in \mathbb{Z}$$

$\therefore$ Identity element is -1

④ INVERSE ELEMENT

For any $a \in \mathbb{Z}$, to find an element $a^{-1} \in \mathbb{Z}$ such that $a * a^{-1} = a^{-1} * a = e$.

Let $a \in \mathbb{Z}$. Consider $a * a^{-1} = e$

$$\Rightarrow a + a^{-1} + 1 = -1$$

$$\Rightarrow a^{-1} = -a - 2 \in \mathbb{Z} \text{ since } a \in \mathbb{Z}.$$

$\therefore$ for any $a \in \mathbb{Z}$, $-a - 2$ is its inverse

$\therefore \mathbb{Z}$ forms a group under $*$.

PROPERTIES OF GROUPS

PROPERTY ①

If every element in a group is its own inverse, then the group must be abelian (OR) if $a^2 = e$ with $a \neq e$, then G is abelian.

Proof:

Let $(G, *)$ be a group. Let $a, b \in G$.

Since every element of G is its own inverse,

$$a = a^{-1} \text{ --- ① and } b = b^{-1} \text{ --- ②}$$

Since $a, b \in G$, $a * b \in G$ ($\because G$ satisfies closure property)

$$\begin{aligned} \therefore a * b &= (a * b)^{-1} \\ &= b^{-1} * a^{-1} \\ &= b * a \text{ (from ① and ②)} \end{aligned}$$

$\therefore$ for any $a, b \in G$, $a * b = b * a$

$\therefore G$ is abelian.

Note:

For any $a, b \in G$, where $(G, *)$ is a group,

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

Proof:

Since $a, b \in G$, $a * b \in G$ ($\because G$ satisfies closure)

To prove $(a * b)^{-1} = b^{-1} * a^{-1}$ (inverse of $a * b$ is $b^{-1} * a^{-1}$)

we need to prove $(a * b) * (b^{-1} * a^{-1}) = e$.

$$\begin{aligned} \text{LHS} &= (a * b) * (b^{-1} * a^{-1}) \\ &= a * (b * b^{-1}) * a^{-1} \quad (\text{Associative property}) \\ &= a * e * a^{-1} \\ &= a * a^{-1} \\ &= e = \text{RHS.} \end{aligned}$$

PROPERTY ②

If G is a group of even order, then it has an element $a \neq e$ such that $a^2 = e$.

Proof:

Since G is a group of even order, let $O(G) = 2n$.

Omitting e , there are $(2n-1)$ elements in G . If we pair off these $(2n-1)$ elements with their inverses, we will be left with only one element say $a \neq e$ which is its own inverse. i.e. $a = a^{-1}$

$$\Rightarrow a * a = a * a^{-1}$$

$$\Rightarrow a^2 = e$$

PROPERTY ③

In a group G , the left and right cancellation laws hold.

Left Cancellation Law

For any $a, b, c \in G$, $a * b = a * c \Rightarrow b = c$

Proof:

Since G is a group, $a \in G \Rightarrow a^{-1} \in G$.

Consider $a * b = a * c$

Premultiply by a^{-1} on both sides,

$$\begin{aligned} a^{-1} * (a * b) &= a^{-1} * (a * c) \\ \Rightarrow (a^{-1} * a) * b &= (a^{-1} * a) * c \quad (\text{Associative property}) \end{aligned}$$

$$\Rightarrow e * b = e * c$$

$$\Rightarrow b = c$$

Right Cancellation Law

For any $a, b, c \in G$, $b * a = c * a \Rightarrow b = c$

Proof:

Assume $b * a = c * a$

Post multiply by a^{-1} on both sides,

$$(b * a) * a^{-1} = (c * a) * a^{-1}$$

$$\Rightarrow b * (a * a^{-1}) = c * (a * a^{-1}) \text{ (Associative property)}$$

$$\Rightarrow b * e = c * e$$

$$\Rightarrow b = c$$

PROPERTY ④

The identity element of a group is unique.

(OR)

If $(G, *)$ is a group and 'e' is an identity element of G then no other element of G is an identity element of G.

Proof:

Let e_1 and e_2 be identity elements of G.

To prove that identity element is unique, we need to P.T. $e_1 = e_2$.

Let $a \in G$. Since e_1 is an identity element of G,

$$a * e_1 = e_1 * a = a \text{ --- ①}$$

Since e_2 is an identity element of G,

$$a * e_2 = e_2 * a = a \text{ --- ②}$$

From ① and ②, $a * e_1 = a * e_2$

$$\Rightarrow e_1 = e_2 \text{ (By Left Cancellation Law)}$$

PROPERTY ⑤

The inverse of any element in a group G is unique.

Proof:

Let e be the identity element of G.

Let $a \in G$. Let b and c be 2 inverses of a.

Since b is the inverse of a, $a * b = b * a = e$ --- ①

Since c is the inverse of a, $a * c = c * a = e$ --- ②

From ① and ②, $a * b = a * c$

$$\Rightarrow b = c \text{ (By Left Cancellation Law)}$$

$\therefore$ Inverse of any element is unique.

PROPERTY ⑥

A group cannot have any element which is idempotent other than the identity element.

Proof:

Let G be a group and let $a \in G$.

Let us assume that a is idempotent.

$$\text{Then } a * a = a \text{ — ①}$$

Let e be the identity element of G .

$$\text{Then } a * e = e * a = a \text{ — ②}$$

$$\text{From ① and ②, } a * a = a * e$$

$$\Rightarrow a = e \text{ (By Left Cancellation Law)}$$

$\therefore$ Identity element is the only idempotent element.

PROPERTY ⑦

If a is an element of a group G , then $(a^{-1})^{-1} = a$.

i.e. If inverse of a is a^{-1} , then the inverse of a^{-1} is a .

Proof:

Let $a \in G$. Then its inverse $a^{-1} \in G$, such that

$$a * a^{-1} = a^{-1} * a = e \text{ — ①}$$

Since $a^{-1} \in G$, its inverse $(a^{-1})^{-1} \in G$ such that

$$a^{-1} * (a^{-1})^{-1} = (a^{-1})^{-1} * a^{-1} = e \text{ — ②}$$

From ① and ②,

$$a * a^{-1} = (a^{-1})^{-1} * a^{-1}$$

$$\Rightarrow a = (a^{-1})^{-1} \text{ (By right cancellation law)}$$

PROPERTY ⑧

If a has inverse b and b has inverse c , then $a = c$.

Proof:

$$a \text{ has inverse } b \Rightarrow a * b = b * a = e \text{ — ①}$$

$$b \text{ has inverse } c \Rightarrow b * c = c * b = e \text{ — ②}$$

$$\text{From ① and ②, } a * b = c * b$$

$$\Rightarrow a = c \text{ (By right cancellation law)}$$

PROPERTY ⑨

A group G is abelian iff $(a*b)^2 = a^2 * b^2$.

Proof:

First we assume that G is abelian.

$$\text{To P.T. } (a*b)^2 = a^2 * b^2$$

Since G is abelian, $a*b = b*a$ for any $a, b \in G$

$$\begin{aligned} \text{LHS} &= (a*b)^2 \\ &= (a*b) * (a*b) \\ &= a * (b*a) * b \\ &= a * (a*b) * b \quad (\because G \text{ is abelian}) \\ &= (a*a) * (b*b) \\ &= a^2 * b^2 = \text{RHS} \end{aligned}$$

Conversely, assume $(a*b)^2 = a^2 * b^2$. To P.T. G is abelian.

$$\begin{aligned} (a*b)^2 &= a^2 * b^2 \\ \Rightarrow (a*b) * (a*b) &= (a*a) * (b*b) \\ \Rightarrow a * (b*a) * b &= a * (a*b) * b \\ \Rightarrow b * a &= a * b \quad (\text{By left and right cancellation laws}) \\ \therefore G &\text{ is abelian} \end{aligned}$$

PROPERTY ⑩

If $(G, *)$ is an abelian group, then for all $a, b \in G$

$$(a*b)^n = a^n * b^n.$$

Proof:

We prove this result by the method of mathematical induction

Step ①

Let $n=1$. $(a*b)^1 = a^1 * b^1$ which is true.

$\therefore$ the statement is true for $n=1$

Step ②

Assume the statement is true for $n=k$.

$$\text{i.e. assume } (a*b)^k = a^k * b^k$$

Step ③

To P.T. the statement is true for $n=k+1$

$$\text{i.e. to P.T. } (a*b)^{k+1} = a^{k+1} * b^{k+1}$$

$$\begin{aligned} \text{LHS} &= (a*b)^{k+1} \\ &= (a*b)^k * (a*b) \\ &= (a^k * b^k) * (a*b) \quad (\text{from step ②}) \\ &= (a^k * b^k) * (b*a) \quad (\because G \text{ is abelian}) \end{aligned}$$

$$\begin{aligned}
&= a^k * (b^k * b) * a \quad (\because G \text{ satisfies associative property) \\
&= a^k * (b^{k+1} * a) \\
&= a^k * (a * b^{k+1}) \quad (\because G \text{ is abelian) } \\
&= (a^k * a) * b^{k+1} \quad (\because G \text{ satisfies associative property) } \\
&= a^{k+1} * b^{k+1} \\
&= \text{RHS.}
\end{aligned}$$

Thus the statement is true for all n .

SUBGROUPS

DEFINITION - SUBGROUP

Let G be a group and H a nonempty subset of G . Then H is a subgroup of G if H itself is a group under the same operation in G .

i.e. Let $(G, *)$ be a group and H be a nonempty subset of G . Then $(H, *)$ is a subgroup of G if

- ① For any $a, b \in H$, $a * b \in H$ (closure property)
- ② For any $a, b, c \in H$, $a * (b * c) = (a * b) * c$ (Associative property)
- ③ $e \in H$ where e is the identity element of G (Existence of Identity)
- ④ For any $a \in H$, $a^{-1} \in H$ (Existence of Inverse)

Eg
 $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +)$
 $(2\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$

Note:

For any group G , the subgroups G and $\{e\}$ are called improper or trivial subgroups and all other subgroups are called proper or non-trivial subgroups.

THEOREM ①

A nonempty subset H of a group G is a subgroup of G iff for any pair of elements $a, b \in H$, $a * b^{-1} \in H$

Proof:

Assume that H is a subgroup of G .

To P.T. for any $a, b \in H$, $a * b^{-1} \in H$

Let $a, b \in H \Rightarrow a, b^{-1} \in H$ ($\because H$ is a subgroup, $b \in H \Rightarrow b^{-1} \in H$)
 $\Rightarrow a * b^{-1} \in H$ ($\because H$ satisfies closure property)

CONVERSELY,

Assume $a, b \in H \Rightarrow a * b^{-1} \in H$ for any $a, b \in H$ — ①

To P.T. H is a subgroup of G , we have to P.T. H itself is a group under $*$.

① ASSOCIATIVE PROPERTY

Since G satisfies associative property and $H \subseteq G$, H also satisfies associative property.

② IDENTITY ELEMENT

Since $H \neq \emptyset$, there exists an element $a \in H$.

$a, a \in H \Rightarrow a * a^{-1} \in H$ (from ①)

$\Rightarrow e \in H$

$\therefore$ Identity element exists in H .

③ INVERSE ELEMENT

We know that $e \in H$. Let $a \in H$

$e, a \in H \Rightarrow e * a^{-1} \in H$ (from ①)

$\Rightarrow a^{-1} \in H$

$\therefore$ Inverse exists for every $a \in H$

④ CLOSURE PROPERTY

Let $a, b \in H$. Then $a, b^{-1} \in H$ (since inverse exists in H)

$\Rightarrow a * (b^{-1})^{-1} \in H$ (from ①)

$\Rightarrow a * b \in H$

$\therefore H$ satisfies closure property.

Thus H is a subgroup of G .

THEOREM ②

The intersection of any two subgroups of a group is also a subgroup of the group, i.e. If H_1 and H_2 are subgroups of a group G , then $H_1 \cap H_2$ is also a subgroup of G .

Proof:

Since H_1 and H_2 are subgroups of G , $e \in H_1$ and $e \in H_2$
 $\Rightarrow e \in H_1 \cap H_2 \quad \therefore H_1 \cap H_2 \neq \emptyset$

To P.T. $H_1 \cap H_2$ is a subgroup of G .

By Theorem ①, it is enough to P.T. $a, b \in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2$

Let $a, b \in H_1 \cap H_2$

$\Rightarrow a, b \in H_1$ and $a, b \in H_2$

$\Rightarrow a * b^{-1} \in H_1$ and $a * b^{-1} \in H_2$ ($\because H_1$ and H_2 are subgroups of G ,
by Theorem ①)

$\Rightarrow a * b^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ is a subgroup of G .

NOTE:

The union of two subgroups of a group G need not be a subgroup of G .

Consider the group $(\mathbb{Z}, +)$

$$2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

$$3\mathbb{Z} = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$$

$(2\mathbb{Z}, +)$ and $(3\mathbb{Z}, +)$ are subgroups of $(\mathbb{Z}, +)$ but
 $(2\mathbb{Z} \cup 3\mathbb{Z}, +)$ is not a subgroup of $(\mathbb{Z}, +)$ since closure
property is not satisfied.

$$2\mathbb{Z} \cup 3\mathbb{Z} = \{\dots, -6, -4, -3, -2, 0, 2, 3, 4, 6, \dots\}$$

For eg., $2, 3 \in 2\mathbb{Z} \cup 3\mathbb{Z}$ but $2+3=5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$

DEFINITION - ORDER OF AN ELEMENT

Let G be a group and let $a \in G$. If there is a least positive integer n such that $a^n = e$, then n is called the order of the element a . If no such integer exists, then a is said to be infinite order.

Eg:

- ① In $(\mathbb{R} - \{0\}, \times)$, the order of -1 is 2
- ② In $(\mathbb{C} - \{0\}, \times)$, the order of $-i$ is 4

DEFINITION - CYCLIC GROUP

Let $(G, *)$ be a group. Then G is said to be cyclic if there exists an element $a \in G$ such that every element $x \in G$ can be expressed as an integral power of a ,

i.e. $x = a^n$ where n is an integer. Here a is called (10)
the generator of the group and we denote it as
 $G = \langle a \rangle$ or $G = \langle a \rangle$

THEOREM (3)

Every cyclic group is abelian.

Proof:

Let $(G, *)$ be a cyclic group generated by an
element $a \in G$, i.e. $G = \langle a \rangle$

Let $x, y \in G$. Then $x = a^m$ and $y = a^n$ where m and n are
integers.

$$\begin{aligned} \text{Consider } x * y &= a^m * a^n \\ &= a^{m+n} \\ &= a^{n+m} \\ &= a^n * a^m \\ &= y * x \text{ for any } x, y \in G. \end{aligned}$$

$\therefore G$ is abelian.

NOTE:

The converse is not true, i.e. Every abelian group is not
cyclic.

THEOREM (4)

Every subgroup of a cyclic group is cyclic.

Proof:

Let G be a cyclic group generated by an element ' a '
and let H be a subgroup of G . To P.T. H is cyclic.

Since H is a subgroup of G , every element of H is of the
form a^n for some $n \in \mathbb{Z}$.

If $a^r \in H$ then $(a^r)^{-1}$ i.e. $a^{-r} \in H$ since H is a subgroup.

Now either r or $-r$ is a positive integer. $\therefore H$ contains
elements with positive indices. Let m be the least positive
integer such that $a^m \in H$.

Claim: a^m is a generator of H .

Let $b \in H$. Then $b = a^n$ for some $n \in \mathbb{Z}$.
 Since m and n are integers, by division algorithm there exist integers q and r such that
 $n = mq + r$ where $0 \leq r < m$.

$$\text{Consider } b = a^n = a^{mq+r} = a^{mq} * a^r = (a^m)^q * a^r \\ \Rightarrow a^n = (a^m)^q * a^r$$

Premultiplying by $(a^m)^{-q}$ on both sides,

$$(a^m)^{-q} * a^n = (a^m)^{-q} * (a^m)^q * a^r = a^r \quad \text{--- (1)}$$

Since $a^m \in H$, $(a^m)^{-q} \in H$ ($\because H$ is a subgroup of G)

Also $a^n \in H \therefore (a^m)^{-q} * a^n \in H$ ($\because H$ satisfies closure)
 $\Rightarrow a^r \in H$ (from (1))

If $r \neq 0$, then $r < m$ and $a^r \in H$ contradicts the fact that m is the least positive integer such that $a^m \in H$. $\therefore r = 0$.

$\therefore b = a^n = a^{mq} = (a^m)^q$
 $\Rightarrow b$ is an integral power of a^m .

$\Rightarrow H = \langle a^m \rangle$
 Hence H is a cyclic subgroup of G .

COSETS

DEFINITION

Let $(H, *)$ be a subgroup of a group $(G, *)$.

Let $a \in G$. Then the set $aH = \{a * h / h \in H\}$ is called the LEFT COSET of H in G determined by the element 'a'.

The RIGHT COSET is defined as $Ha = \{h * a / h \in H\}$.

Obviously Ha and aH are subsets of G .

EXAMPLES

1. Let $G = \{1, a, a^2, a^3\}$ with $a^4 = 1$ and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H .

Soln.

To find the right cosets and left cosets of H in G .

Right Cosets

$$H1 = \{1, a^2\} = H$$

$$Ha = \{a, a^3\}$$

$$Ha^2 = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$Ha^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$$

$\therefore$ The distinct right cosets are H and Ha .

Left Cosets

$$1H = \{1, a^2\} = H$$

$$aH = \{a, a^3\}$$

$$a^2H = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$a^3H = \{a^3, a^5\} = \{a^3, a\} = aH$$

$\therefore$ The distinct left cosets are H and aH .

2. Find the left cosets of $\{[0], [2]\}$ in the group $(\mathbb{Z}_4, +_4)$
Soln.

$\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ is a group under $+_4$

Clearly $H = \{[0], [2]\}$ is a subgroup of $\mathbb{Z}_4$ under $+_4$.

Left Cosets of H

$$[0] + H = \{[0] + [0], [0] + [2]\} = \{[0], [2]\}$$

$$[1] + H = \{[1] + [0], [1] + [2]\} = \{[1], [3]\}$$

$$[2] + H = \{[2] + [0], [2] + [2]\} = \{[2], [0]\} = H$$

$$[3] + H = \{[3] + [0], [3] + [2]\} = \{[3], [1]\} = [1] + H$$

$\therefore$ Distinct left cosets of H in $\mathbb{Z}_4$ are H and $[1] + H$.

NOTE:

1. The left cosets (or right cosets) are mutually disjoint and their union is G .

2. Since H is a subgroup of G , $e \in H$. Thus for any $a \in G$, $a = a * e \in aH$ and $a = e * a \in Ha$.

$\therefore$ Every element of G belongs to a left (or right) coset of H .

3. H itself is a left and a right coset, since $eH = H$ and $He = H$.

4. If G is abelian, then there is no distinction between the left cosets and right cosets of H .

i.e. $aH = Ha \forall a \in G$ if G is abelian.

5. If $a \in H$, then $aH = Ha = H$.

DEFINITIONS

1. CONGRUENCE MODULO H

Let G be a group and H a subgroup of G . For $a, b \in G$ we say that a is congruent to b modulo H written as $a \equiv b \pmod{H}$ if $b^{-1} * a \in H$

2. EQUIVALENCE CLASS

For any $a \in G$, the equivalence class determined by 'a' is $[a] = \{x \in G / x \equiv a \pmod{H}\} = \{x \in G / a^{-1} * x \in H\}$

THEOREM (5)

Any two right or left cosets of H in G are either identical or disjoint.

Proof:

Let $a \in G$. We know that $[a] = \{x \in G / x \equiv a \pmod{H}\}$

We first P.T. $[a] = aH$.

Let $x \in aH \Rightarrow x = a * h$ for some $h \in H$

$$\Rightarrow a^{-1} * x = h$$

$$\Rightarrow a^{-1} * x \in H$$

$$\Rightarrow x \equiv a \pmod{H}$$

$$\Rightarrow x \in [a]$$

$$\therefore aH \subseteq [a] \text{ --- (1)}$$

By retracing the steps, we can P.T. $[a] \subseteq aH$ --- (2)

From (1) and (2), $[a] = aH$

We know that any two equivalence classes are either identical or disjoint. $\therefore$ Any two left cosets are either identical or disjoint.

THEOREM (6)

There is a one to one correspondence between any two left cosets of H in G .

Proof:

One to one correspondence means 1-1 and onto.

Let aH and bH be any two left cosets of H in G .

Define a map $f: aH \rightarrow bH$ by $f(a * h) = b * h$, for $h \in H$.

- (i) f is clearly onto since for every $b * h \in bH$ where $h \in H$ we can find an element $a * h \in aH$ for this h such that $f(a * h) = b * h$.
- (ii) To prove f is 1-1, we have to prove that
 $f(a * h_1) = f(a * h_2) \Rightarrow a * h_1 = a * h_2$
 Assume $f(a * h_1) = f(a * h_2)$
 $\Rightarrow b * h_1 = b * h_2$
 $\Rightarrow h_1 = h_2$ (By left cancellation law)
 $\Rightarrow a * h_1 = a * h_2$ (Premultiplying by a on both sides)
 $\therefore f$ is 1-1.

COROLLARY

Let H be a subgroup of a group G . For any $a \in G$, Ha is any right coset of H in G . Since H itself is a right coset, there exists a 1-1 correspondence between the elements of H and Ha . Thus H and any right coset of H in G will have the same number of elements.

THEOREM (7)

LAGRANGE'S THEOREM

The order of a subgroup of a finite group divides the order of the group.

Proof:

Let $(G, *)$ be a finite group of order $n \Rightarrow o(G) = n$

Let $(H, *)$ be a subgroup of $(G, *)$ of order $m \Rightarrow o(H) = m$

To P.T. $o(H) \mid o(G)$ i.e. to P.T. $m \mid n$.

i.e. to P.T. $n = km$ for some integer k .

Let $h_1, h_2, \dots, h_m$ be the elements of H . For any $a \in G$, the right coset Ha of H in G is defined by

$Ha = \{h_1 * a, h_2 * a, \dots, h_m * a\}$. By corollary to Theorem (6), since H has m elements, each right coset of H in G has m distinct elements.

Since G is a finite group, the number of distinct

right cosets of H in G will be finite, say k .

Since H is a subgroup of G , $e \in H$.

$\therefore$ For any $a \in G$, $a = e * a \in Ha$. Thus every element of G appears in atleast one right coset.

$\therefore$ The union of all these k distinct right cosets of H in G is equal to G .

If $Ha_1, Ha_2, \dots, Ha_k$ are the k distinct right cosets of H in G , then

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$$

$$\Rightarrow O(G) = O(Ha_1 \cup Ha_2 \cup \dots \cup Ha_k)$$

$$\Rightarrow n = O(Ha_1) + O(Ha_2) + \dots + O(Ha_k) \quad (\because Ha_1, Ha_2, \dots, Ha_k \text{ are all distinct})$$
$$= m + m + \dots + m, k \text{ times}$$

$$\Rightarrow n = km$$

THEOREM (8)

Let H be a subgroup of G . Then the set of all right cosets of H in G forms a partition of G .

Proof:

- (i) For any $a \in G$, $a \in Ha$. Hence any right coset is nonempty
- (ii) Every element of G appears in atleast one right coset of H in G . $\therefore$ Union of all the right cosets of H in G will be equal to G .
- (iii) Any two right cosets of H in G are either identical or disjoint. (By Theorem (5))

From the above, we see that the set of all right cosets of H in G forms a partition of G .

DEFINITION - GROUP HOMOMORPHISM

Let $(G, *)$ and (G', Δ) be two groups. A mapping $f: (G, *) \rightarrow (G', \Delta)$ is called a group homomorphism if for any $a, b \in G$,

$$f(a * b) = f(a) \Delta f(b)$$